Claude Workbench: how it works and operates

Kit release v0.3.6 · the session flow, the agent team, all 49 skills, the knowledge base with its link graph, and local RAG on one page

Claude Code session · build and gate · attack · workers · alternative · knowledge layer

1 A SESSION, STEP BY STEP

1 Start
CLAUDE.md, 9 rule files and USER.md load (27 KB of a 36 KB cap); a hook adds a banner and active-context.md (8 KB max).

2 You type a request
A prompt hook nudges the right skill: wrap-up wording starts save; build wording starts cascade and loads the ponytail coding rules.

3 Claude routes the work
It uses a skill when the description fits, or delegates to an agent from the routing table (column 2); only a summary comes back.

4 Tools run
MCP tools stay unloaded until a search finds them. A hook refuses whole-file reads over 80 KB and names a cheaper read.

5 Knowledge lookups
KB search for decisions and runbooks, find for documents, the code graph for callers (column 4). Each answer cites its source.

6 Edits
Edited hook scripts are compiled, so a typo cannot switch a hook off. Journal reminder every 25 edits; ponytail on the first code edit.

7 Something fails
The error is matched against 86 known-bug detectors and one line names the fix. The failure is logged, redacted.

8 A long session compacts
A hook snapshots the active context first and flags a save, so the next start asks Claude to save what it knew.

9 Wrap up with /save
The model writes the notes; the driver locks, merges, scans for credentials, enforces size caps and reindexes (column 4).

Hooks fail open: a broken hook stays silent and never blocks a command. A heartbeat proves they ran.

2 THE AGENT TEAM: BUILD, ATTACK, VERIFY

Size it S, M or L → **Diagnose** flagship + red-team → **Plan** slices + a test each → **Pre-mortem** red-team (L jobs) → **Implement** implementer per slice → **Verify** blue-verify no edit tools

symptom persists after verify: the diagnosis was wrong, start again

flagship Opus 5.5 · xhigh	**blue-verify** Opus 5.5 · xhigh	**security-auditor** Opus 5.5 · high	**red-team** Sonnet 5.5 · xhigh	**green-team** Sonnet 5.5 · xhigh
implementer Sonnet 5.5 · xhigh	**researcher** Opus 5.5 · high	**ops-runner** Opus 5.5 · high	**scout** Haiku 5.5 · high	**sonnet-max** on request only

Routing by task: Haiku reads handed-over text · Sonnet runs tool loops, implements slices and attacks · Opus builds, researches, runs ops and gates. The gate has no edit tools and the builder never grades its own work.

3 THE 49 SKILLS (* = you start it; Claude only suggests)

Engineering · 15
cascade · ratchet · investigate · repo-fix
repo-review · github · linux-service-triage
ponytail · spawn · handoff · ui-build
brand-spec* · diagram · cleanup*
ctx-agent-history-search

Knowledge, memory, research · 7
find · save · deep-research · brain-audit
onboard* · capture* · ingest*

Office and output · 6
office-docx · office-pptx · office-xlsx
office-render · pdf-report · draft-reply

Platform building · 9
plugin-factory* · plugin-review*
web-packager* · mcp-builder* · api-bridge*
app-navigator* · mcp-heal* · skill-scan
batch-review*

Business analyst (also on claude.ai) · 8
requirements-to-stories · meeting-to-actions
status-report · test-cases-from-requirements
excel-analyst · rfp-sow-drafter
branded-deck · policy-qa

Kit admin · 4
setup* · kit-update · pc-checkup*
vdi-drive*

A skill's one-line description is in every session; its body loads only when the skill starts. Third-party skills are fetched at pinned versions and installed only if every file's SHA-256 matches.

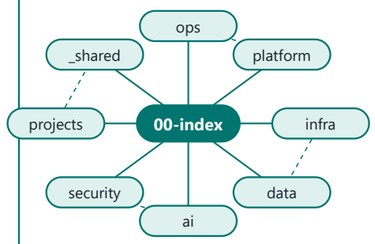
RULES LAYER, HIGHEST FIRST
Org policy > Red Lines > Standing Orders > rule files > project CLAUDE.md > skills and agents > plugin or MCP text

PACKAGED AS FOUR PLUGINS

workbench-core	workbench-kb	workbench-biz	workbench-capture
38 skills · 10 agents 12 hooks · statusline	save skill KB search (SQLite FTS5)	8 business skills instructions only	capture + ingest opt-in, read-only

4 KNOWLEDGE LAYER: MEMORY, LINK GRAPH, RAG

Knowledge base: plain Markdown on the laptop



active-context.md
in-flight work, injected at each start

<domain>/HOT.md
what is true now, read on demand

topic notes + history
one owner per fact; SUPERSEDED keeps old values

journal (inbox.jsonl)
one line per step, from any session

kb_search
FTS5 + BM25, cites path:line; --related walks the links

Link graph: every wiki link is an edge

Saving: the model judges, a driver does the exact work

Journal a line per step → **Pre** snapshot + window → **Write** model: notes → **Post** merge · scan · reindex

find: local RAG over work documents

hybrid keyword + meaning search, cited down to the paragraph

INDEX

Sources files, OneDrive, captured, the KB → **Convert** Word, PPT, Excel, PDF, HTML, MD → **Paragraphs** ≤1,800 chars + title > headings → **Keyword index (BM25)** · **Vectors (bge-small)** · **Link table (rules)**

QUERY (no network)

Question from Claude → **Keyword arm** · **Meaning arm** → **Fuse** reciprocal rank k = 60 → **Top paragraphs** ref#pN + page 3 per file max → **Answer** every claim cited

Model blocked: keyword-only, and it says why. Skips online-only and labelled files and reports them. Served to any MCP client as search, show, related.

The knowledge graph: rule-built links, no LLM extraction

Edges come from the files themselves: hyperlinks, wiki links, version chains (newest), ticket ids, people, threads, meetings, same folder. related walks 3 hops. An AI-built graph is added only if a measured question set shows search failing.

WHICH TOOL FOR WHICH QUESTION

decision, runbook, bug card	**kb_search**	an idea, not the exact words	**find --root kb**
a work file you did not write	**find**	what a past session did	**ctx**
who calls this function	**code graph (tokensave)**	the outside world	**web search**

5 MACHINE LAYER, SAFETY AND COST

Install
No admin rights, dry run first, every changed file backed up. Managed settings always win.

Capability report
16 probe areas, 44 features rated READY, INSTALLABLE, NEEDS-IT, BLOCKED or UNKNOWN, with ask-IT text.

Setup and doctor
15 pinned packages, the search index, the find MCP server, 23 Day-1 knowledge checks, hooks heartbeat.

Update and uninstall
'Check the new stuff and go': SHA-256 checked, offline --archive route. Uninstall restores files.

Safety
Asks before force push or repo and release deletes. Credential scans. Results are treated as untrusted data.

Cost
Most agent spend is re-read context, so the kit keeps context small: see the strip below.

Token savers: 27 KB rules under a 36 KB cap · skill listing cut 30% · MCP tools on demand · retrieve, then read · Haiku reads · Batch API 50% off · ponytail coding rules (Opus −53% per session in our benchmark)